

Shapes and Angles

04



Shapes are the building blocks of objects. We can create many objects by combining shapes, just as it is done in this tangram.

Nice. Can you show me how you made a tangram?



Everything we see around us is made up of a structure. An object's basic structure is made up of one or more shapes. We use shapes to identify objects.

The shape of an object is its form. For example: the shape of a book is usually a rectangle and the shape of a pizza slice is triangle. The geometric shape of an object can determine how big or small it is as compared to the other objects around it.

4. Shapes and Angles

There are two main types of shapes: open and closed.



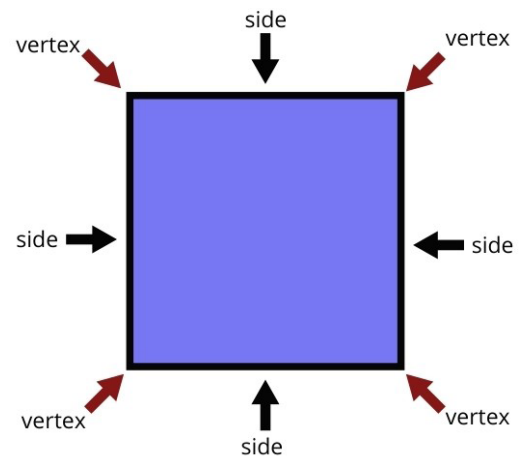
Open



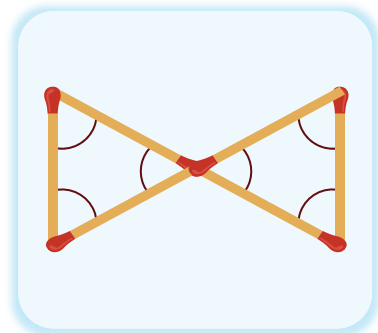
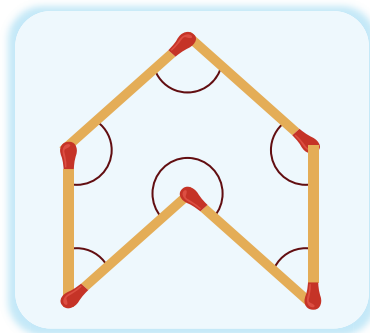
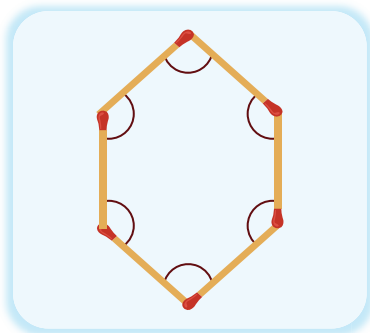
Closed

A shape consists of sides and vertices.

- Sides are the lines that make up a shape.
- A vertex is the pointed corner where two sides meet.



An angle is formed when two sides meet at a vertex.



Activity

Use shapes to create a tangram puzzle

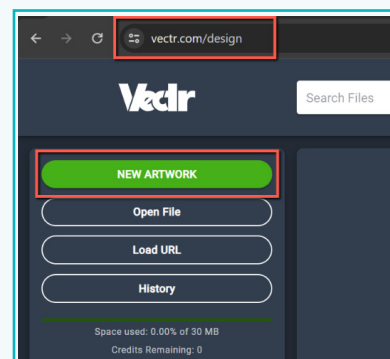
A tangram is a type of ancient Chinese puzzle. It has 7 specific pieces which fit together perfectly to form a **square**. The 7 pieces consist of 2 large **triangles**, 1 medium **triangle**, 2 small **triangles**, 1 **square**, and 1 **parallelogram**. A Parallelogram is a flat shape where the sides opposite to each other are parallel and equal in length.

A tangram's shapes can be rearranged to create different objects. You can create new characters by trying different combinations of shapes using reference images from puzzle books or by using your own imagination.



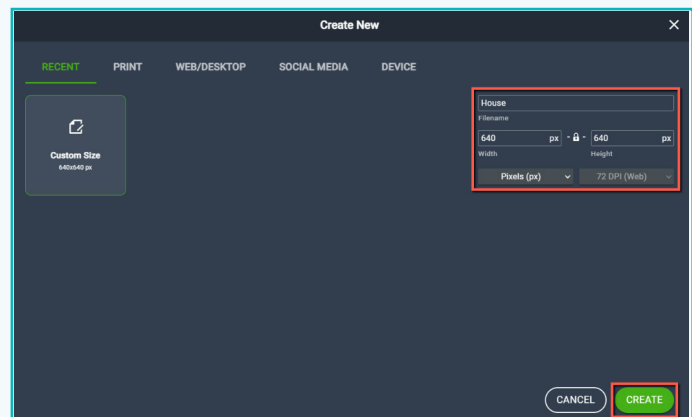
To create the tangram

1 Create a new file in vectr.com.



2 Before we start, let's set up the document.

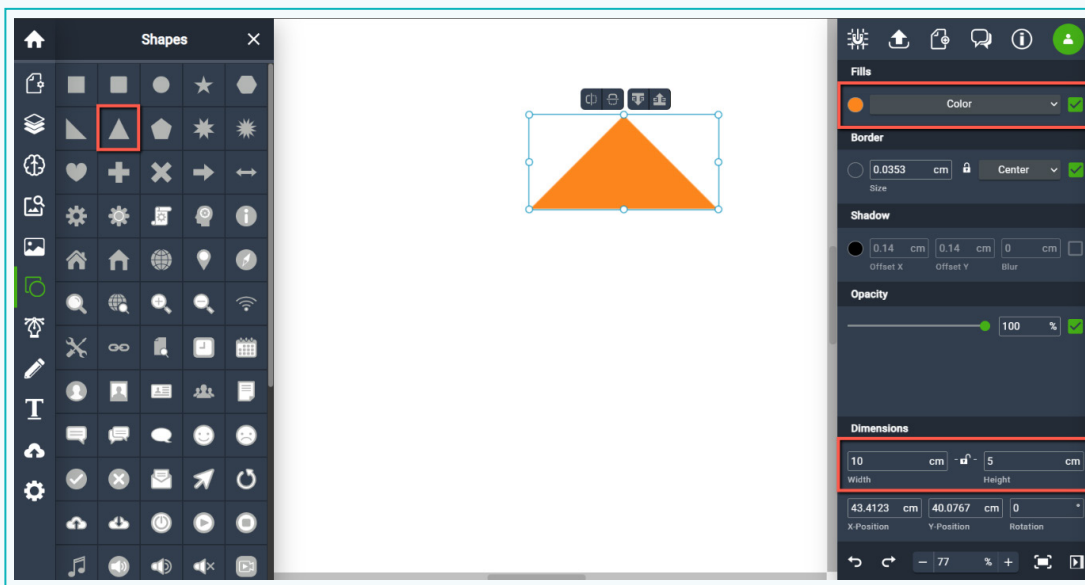
- Add a file name and unit system for the document.
- For this, we will use centimetre.
- Click on Create.



Activity

3 From the More Shapes tool, create a triangle sized 10 cm X 5 cm.

a. Set the colour and dimensions of the triangle from the properties panel as shown.

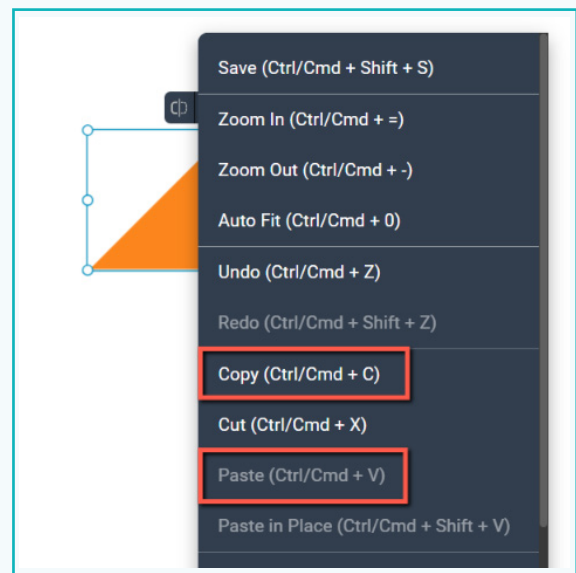


Note:

In Dimensions, make sure the link between width and height is not checked. When the link is checked, the object maintains the aspect ratio when size changes.

4 To make another triangle of the same size, we will use the duplicate function.

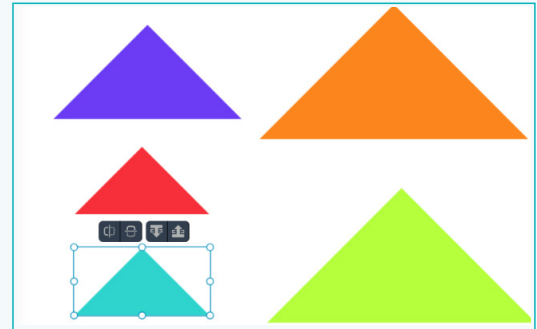
- Right click on the triangle and select Copy.
- Again right click to select paste.



Activity

5 Similarly, create the rest of the triangles with the following sizes:

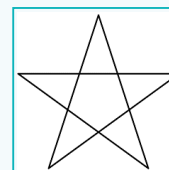
- 1 medium triangle: 7 cm X 3.5 cm.
- 2 small triangles: 5 cm X 2.5 cm.



6 Now, create 1 square sized 3.5cm.

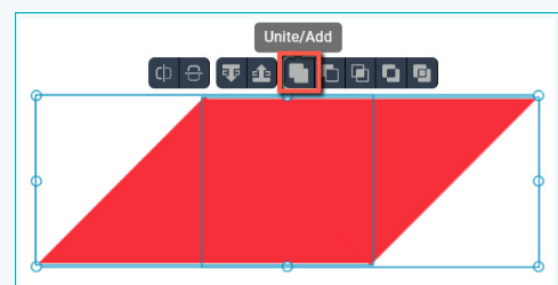
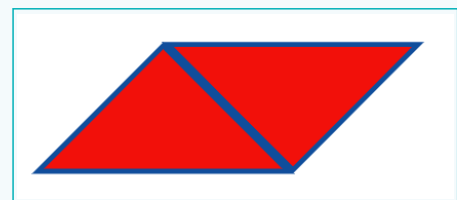


- A simple shape has only one boundary. Its boundary does not cross over itself. A square is a simple shape.
- A complex shape is formed with 2 or more shapes. The boundary of a complex shape can cross over itself. For example: A pentagram (a 5-sided shape).



7 To create a parallelogram:

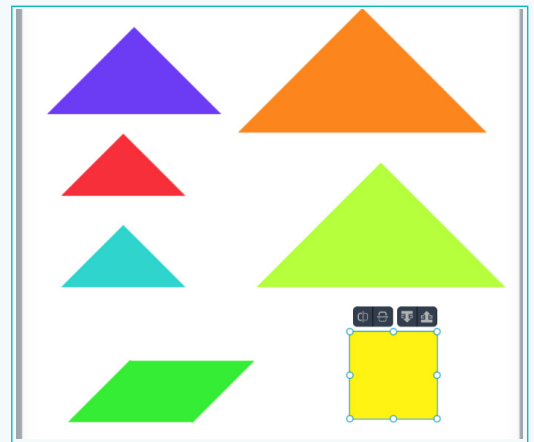
- We will create 2 triangles and place them as shown in the image.
- To join the 2 shapes, we will use the unite command.
- Hold down the shift key and select both triangles. Right click and select unite.



Activity

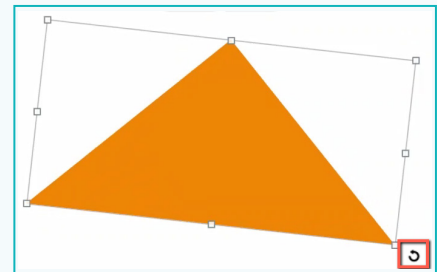
8 Our shapes for the tangram are ready:

- a. 2 large triangles: 10 cm X 5 cm
- b. 1 medium triangle: 7 cm X 3.5 cm
- c. 2 small triangles: 5 cm X 2.5 cm
- d. 1 square: 3.5 cm
- e. 1 parallelogram: 5 cm X 2.5 cm

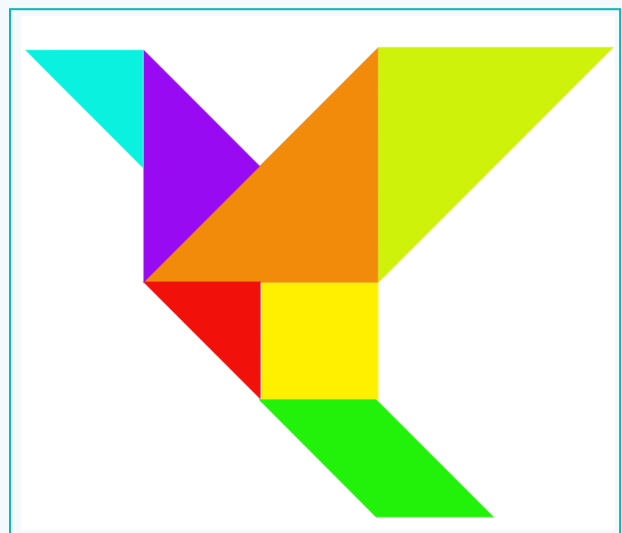


9 Now that our shapes are ready we can rearrange them to create different characters and objects.

- 10** To rotate: Select the shape, hold the cursor at one of the corner points until the rotate cursor appears. Once the cursor is visible, left click and drag to rotate. You can also change the Rotation value in the Properties Panel.

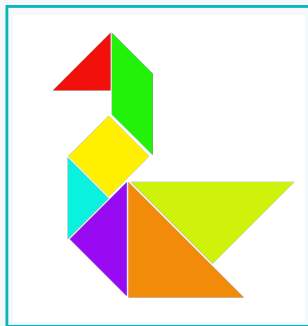


- 11** Rotate shapes to create an object.

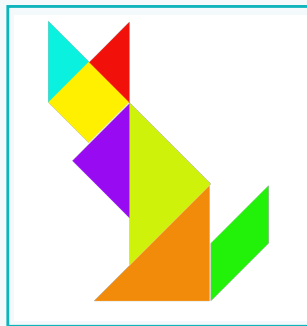


Activity

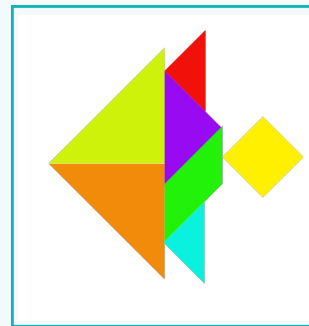
- 12 Try to create the following characters. Save them in .PNG format.



Swan



Cat



Fish

Today we learnt about the ancient Tangram puzzle. It is created by using 7 geometrical shapes. Use your imagination to make as many different combinations as you can. What other objects can you make?

Exercise

Fill in the blanks

- 1 A _____ shape has only one boundary which does not cross over itself.
- 2 _____ is the keyboard shortcut for duplicate.
- 3 The _____ command is used to combine multiple shapes to one.
- 4 The point where edges meet is called a _____.



Tech Flash

- **Shape** : A shape is the form of an object.
- **Properties panel** : The properties panel is used to set the dimension of a shape.
- **Multiple selection** : To hold multiple objects, hold shift and click on the objects to be selected.